Example 4.4 In a parallelogram $ABCD$, P and Q are the points on line DB such that $PD = BQ$ show that $APCQ$ is a parallelogram

Solution

$ABCD$ is a parallelogram.

$$OA = OC \text{ and}$$

$$OB = OD \ (\because \text{Diagonals bisect each other})$$

now $$OB + BQ = OD + DP$$

$$OQ = OP \text{ and } OA = OC$$

$APCQ$ is a parallelogram.

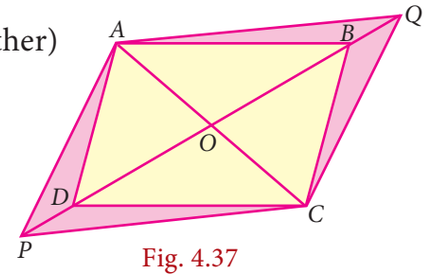


Fig. 4.37

Exercise 4.2

1. The angles of a quadrilateral are in the ratio 2 : 4 : 5 : 7. Find all the angles.
2. In a quadrilateral $ABCD$, $\angle A = 72°$ and $\angle C$ is the supplementary of $\angle A$. The other two angles are $2x–10$ and $x + 4$. Find the value of x and the measure of all the angles.
3. $ABCD$ is a rectangle whose diagonals AC and BD intersect at O. If $\angle OAB = 46°$, find $\angle OBC$
4. The lengths of the diagonals of a Rhombus are 12 cm and 16 cm . Find the side of the rhombus.
5. Show that the bisectors of angles of a parallelogram form a rectangle .
6. If a triangle and a parallelogram lie on the same base and between the same parallels, then prove that the area of the triangle is equal to half of the area of parallelogram.
7. Iron rods a, b, c, d, e, and f are making a design in a bridge as shown in the figure. If $a \parallel b$, c $\parallel$ d , $e \parallel f$, find the marked angles between

   (i) b and c

   (ii) d and e

   (iii d and f

   (iv) c and f

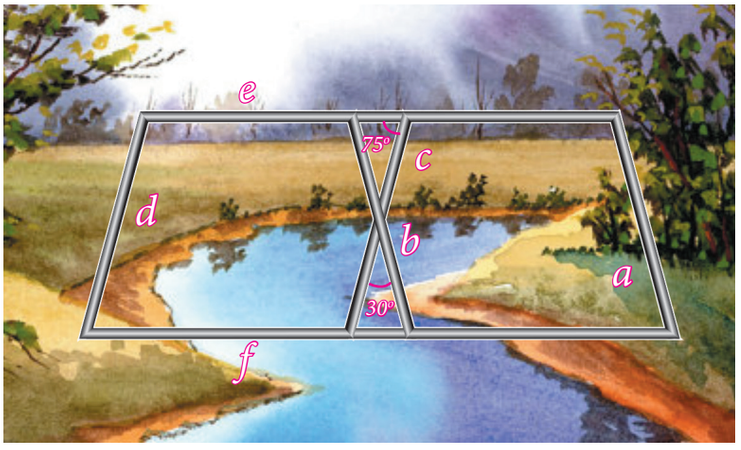


Fig. 4.38

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